Homogeneous Systems

Recall from last class:

- Let A be an $m \times n$ matrix, then
$$A = [\vec{a}_1 \ \dots \ \vec{a}_n], \ \ \vec{a}_j \in \mathbb{R}^m, \ 1 \leq j \leq n.$$
If $x \in \mathbb{R}^n$, then we define
$$A\vec{x} = \vec{a}_1 x_1 + \vec{a}_2 x_2 + \cdots + \vec{a}_n x_n.$$
Thus the matrix equation $A\vec{x} = \vec{b}$ is the same as the vector equation
$$\vec{a}_1 x_1 + \vec{a}_2 x_2 + \cdots + \vec{a}_n x_n = \vec{b}.$$

- $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b} \in \text{Span}(\vec{a}_1, \dots, \vec{a}_n)$ where $\vec{a}_j$ is the j-th column of A.

- A set of vectors $\vec{v}_1, \dots, \vec{v}_p$ in $\mathbb{R}^m$ **spans** $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $\vec{v}_1, \dots, \vec{v}_p$.

- **Theorem** Let A be an $m \times n$ matrix, then the following are equivalent.
  1. For each $\vec{b} \in \mathbb{R}^m$, the matrix equation $A\vec{x} = \vec{b}$ is consistent.
  2. Each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.
  3. The columns of A span $\mathbb{R}^m$.
  4. A has a pivot position in every row.

- **Fact (or Theorem 5 from the book):** If A is a $m \times n$ matrix and $\vec{u}, \vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$, then
  1. $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$
  2. $A(c\vec{u}) = c(A\vec{u})$.

Definition 1. *A linear system is called* **homogeneous** *if it can be written as* $A\vec{x} = \vec{0}$ *where* A *is an* $m \times n$ *matrix,* $\vec{x} \in \mathbb{R}^n$*, and* $\vec{0}$ *is the vector of* m *zeros.*

1. Let $A = \begin{bmatrix} 2 & -5 & 8 \\ -2 & -7 & 1 \\ 4 & 2 & 7 \end{bmatrix}$. Find all solutions to $A\vec{x} = \vec{0}$.

Theorem: If $A\vec{x} = \vec{b}$ is consistent for a given $\vec{b}$, and $\vec{p}$ is a solution, then the solution set of $A\vec{x} = \vec{b}$ is the set of vectors of the form $\vec{p} + \vec{v}_h$ where $\vec{v}_h$ is a solution of $A\vec{x} = \vec{0}$.

2. Prove the Theorem.

3. Let $A = \begin{bmatrix} 1 & 0 & 5 \\ -2 & 1 & -6 \\ 0 & 2 & 8 \end{bmatrix}$.

(a) Find all solutions to $A\vec{x} = \vec{0}$.

(b) Let $\vec{b} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$. Describe all solutions to $A\vec{x} = \vec{b}$.

4. Construct a 2 by 2 matrix A such that the solution set to $A\vec{x} = \vec{0}$ is the line in $\mathbb{R}^2$ going through $(0,0)$ and $(4,1)$.

Definition 2. *A set of vectors* $\{\vec{v}_1, \ldots, \vec{v}_p\}$ *in* $\mathbb{R}^n$ *is said to be* **linearly independent** *if*

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution. The set is called **linearly dependent** *if the equation has non-trivial solutions and the equation is called the* **dependence relation**.

5. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Is $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ linearly independent?

Let $A = [\vec{a}_1 \ \ldots \ \vec{a}_n]$, then $\{\vec{a}_1 \ \ldots \ \vec{a}_n\}$ are linearly independent if and only if $A\vec{x} = \vec{0}$ has only the trivial solution.

6. Is a set with only one vector always linearly independent?

7. If $\vec{0} \in \{\vec{v}_1, \ldots, \vec{v}_p\}$, then is the set linearly dependent?

8. If $\vec{0} \in \text{Span}\{\vec{v}_1, \ldots, \vec{v}_p\}$, then the vectors are linearly independent?

9. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$. Are the vectors linearly independent?

Theorem 7: A set $S = \{\vec{v}_1, \ldots, \vec{v}_p\}$ of two or more vectors is linearly dependent if and only if one of the vectors of S is a linearly combination of the others.

Moreover if $\vec{v}_1 \neq \vec{0}$, then some $\vec{v}_j$ for $j > 1$ is a linear combination of $\vec{v}_1, \ldots, \vec{v}_{j-1}$.

10. Prove Theorem 7.

11. Let $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}$

(a) Describe Span$(\vec{u}, \vec{v})$

(b) Prove $\vec{w} \in Span(\vec{u}, \vec{v})$ if and only if $\{\vec{u}, \vec{v}, \vec{w}\}$ is linearly dependent.

Theorem 8: Any set $\{\vec{v}_1, \ldots, \vec{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

12. Prove the theorem.

Homogeneous Systems and Linear Independence – Answers / Solutions

1. The REF for $[A|0]$ (here 0 means the 3×1 column vector $0 \in \mathbb{R}^3$) is:

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{17}{8} & 0 \\ 0 & 1 & -\frac{3}{4} & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

There are two basic variables x_1, x_2 and one free variable x_3. The general solution is

$$x_3 \begin{bmatrix} -\frac{17}{8} \\ \frac{3}{4} \\ 1 \end{bmatrix}$$

2. Consider the two sets $S_1 = \{\vec{x} \mid A\vec{x} = \vec{b}\}$ and $S_2 = \{\vec{p} + \vec{v}_h \mid A\vec{v}_h = 0\}$. The first set S_1 is just the usual solution set of $A\vec{x} = \vec{b}$, and the second set S_2 is the set described in the problem. Our goal is to show that these two things are equal. The way we will do this is by first showing that $S_1 \subset S_2$ (that every element of S_1 is contained in S_2) and $S_2 \subset S_1$ (every element of S_2 is contained in S_1).

$S_1 \subset S_2$: Let $\vec{x} \in S_1$. We also know that $A\vec{p} = \vec{b}$. Therefore, $A(\vec{x} - \vec{p}) = A\vec{x} - A\vec{p} = \vec{b} - \vec{b} = 0$. So let $\vec{v}_h = \vec{x} - \vec{p}$. Then $\vec{p} + \vec{v}_h = \vec{p} + \vec{x} - \vec{p} = \vec{x}$, so $\vec{x} \in S_2$.

$S_2 \subset S_1$: Let $\vec{p} + \vec{v}_h \in S_2$. Apply A to this element: $A(\vec{p} + \vec{v}_h) = A\vec{p} + A\vec{v}_h = \vec{b} + \vec{0} = \vec{b}$, so we have that $\vec{p} + \vec{v}_h \in S_1$ is a solution to $A\vec{x} = \vec{b}$.

Therefore, since both $S_1 \subset S_2$ and $S_2 \subset S_1$, then they must be equal.

3. (a) The REF of the augmented matrix $[A|\vec{0}]$ is:

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

There are two basic variables x_1, x_2 and one free variable x_3. The general solution is:

$$x_3 \begin{bmatrix} -5 \\ -4 \\ 1 \end{bmatrix}$$

(b) The REF of the augmented matrix $[A|\vec{b}]$ is:

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Again, there are two basic variables x_1, x_2 and one free variable x_3. The general solution is:

$$\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -5 \\ -4 \\ 1 \end{bmatrix}$$

4. The line passing through $(0, 0)$ and $(4, 1)$ has equation $y = \frac{1}{4}x$. We can also write this as $x - 4y = 0$. Therefore, the matrix

$$\begin{bmatrix} 1 & -4 \\ 0 & 0 \end{bmatrix}$$

has the correct solution set.